

Contents

1 · The Three Closures, Consolidated	2
2 · The Escape, Quantified — and Re-Adjudicated	5
3 · The Unified Statement and the Two Remaining Walls	10
4 · The Battery Row and the Corpus Dictionary	13
5 · The Corrigendum — the Escape Retracted, the Valley Adjudicated	14
6 · The Two Walls, Certified Symmetric	16
7 · The Tail's Law, the Split Refuted, and the Critical Locus Closed	20
8 · The Regenerated Ledger — the Batteries, the Dictionary, the Residue	23

1 · The Three Closures, Consolidated

Volume XII closed with a ledger of five open links, each with its owner instrument and its named next step. This volume opens with three of them closed. The order came as a single sentence — consolidate the three closures into the unification — and the consolidation is mechanical in the best sense: each closure was already anchored to its battery, its certificate tier already typed, and what remained was to read the three of them into the unified statement as a single picture. The continuum Fisher-Rao/Leray upgrade closes the first link — the (f) bridge's named residual, the honest price of the discrete adjudication. The off-class nc-AAK package closes the second — Open 7.13's first constructive delivery on the canonical witness, together with the escape whose full quantification was that closure's own named residual. The rank-aware synthesis closes the fourth — the physics rung's designed-but-unbuilt instrument. Table 1 states the three closures at the level the batteries proved them; the sections below read each into the unification's spine.

5 to 2 the open ledger's links, before to after	31 task batteries, all machine-anchored	8.2e-9 the escape's certified residual
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The first closure — the continuum upgrade — is the one the unification needed most, because Volume XII's adjudication had priced it honestly: the discrete Lyapunov-cohomology correspondence was proved exactly, and the Fisher-Rao and Leray components of the treatises remained unbridged at the certified level. Task 28 built the bridge on the instance the corpus always tests on: the driven circle diffusion, the continuum limit of Task 24's N-ring. The convergence theorem is the load-bearing wall — the discrete Kolmogorov circulations $N \log(a/b)$ converge to the de Rham class $2 \pi \mu / D$ with measured order 2.000 over $N = 8$ to 8192 — because it makes the discrete correspondence the skeleton of the continuum one rather than a metaphor for it. On top of it: the FKLZ Lyapunov 1-form on the drift flow (non-exact iff μ is not zero), the potential criterion (the periodic Fourier solve exists iff the class vanishes iff $\mu = 0$ — the detailed-balance slice upgraded), the Stein/Green homotopy with its spectral ladder to $-Dk$ squared over $2 \text{ plus } i \mu k$, the Green kernel's antisymmetric part as the class's machine-verified witness, the true Fisher-Rao policy bundle with the H-theorem at $5.7\text{e-}11$ and the replicator identification at $9.4\text{e-}16$, and the two-patch Cech-de Rham machine whose overlap cocycle differs by exactly the class — machine-exact. The viability wall and its discontinuous safe-mode reset carry over to the continuum as the honest frontier: the interior loops' payments decay as ω squared, the crossing loops pay the rate-independent reset lump of about 0.0158, and the ratchet — the state returns, the payment accumulates — is the arrow's monotone account. The Leray page on the environment-loop fibration collapses by degree reasons and its edge homomorphism is the reset payment: the treatise's Leray step, delivered on the instance.

The second closure — the off-class problem — is Open 7.13 on the free cell, and its shape is the corpus's signature move: type the class first, obstruct second, trade off third. The class theorem makes $D_{\text{free}}(2)$ the free Nehari distance — the 2-state WFA class is exactly the free Hankel operators of rank at most 2, both directions constructive to $1e-15$, so the quantity Open 7.13 asks to characterize is intrinsic, with no syntax in it. The window obstruction then proves the constructive theory has genuine off-class content: the cell's cut-web window contains a forced 3×3 minor of determinant exactly minus one, whatever the free entries — no rank-2 completion matches the window, the corner is structurally unkillable, and the optimization is forced into the corner-payment trade-off the abelian cell already certified. The finite-section ladder measured 1.199, 1.232, 1.261 at windows 2, 3, 4 — climbing past the EYM floor of 1 that plain section singular values can never beat (interlacing caps them forever): the converse must see the Hankel structure, not the spectrum. And the escape — the 12-dimensional descent from the abelian optimizer beating the shadow — was the discovery, reported in Task 29 as a $5.7266e-7$ refutation of the shadow-equality conjecture. Chapter 2 of this volume quantifies it further, as ordered, and in doing so re-adjudicates it: the number decomposes, the shadow moves, and the refutation stands at $8.2e-9$ — a seventy-fold correction that the reproduction-first discipline found because it never trusts a prior session's own report of its own numbers.

The third closure — the rank-aware synthesis — completes the physics rung, the docking matrix's fourth link, designed in Volume I's seam analysis and not yet built. Task 30 built it end to end. The rank-across-cuts law is the law the whole corpus was pointing at: the one-ended chain's dense Green's function is rank 1 at every cut (the second singular value below $2e-16$), the Euler-Bernoulli beam's is rank 2 (the transmission state: deflection plus rotation), the grid's is the full cross-section — the cross-rank is the interface's physical state dimension, and the classical transmission conditions are the sufficient statistics. The two scalars are certified exact to $1e-13$; the sequential load-decision dynamic program on the two-scalar state (S, W) with the exact recursion $y_i = S_{\{i-1\}} + i W_i$ agrees exactly with the brute force over all 531,441 load patterns — the naive exponential width replaced by the polynomial lattice. The truncated-statistics sandwich bounds and attains the beam's $M = 1$ error at the EYM floor with the directed witness exact, and the magnitude punchline is the row the dictionary wanted: the banded, decay-thresholded approximation of the dense rank-1 chain pays operator errors of order 100 at every bandwidth while the one-scalar rank-aware interface is exact — rank, not magnitude. The exact rational minimality certificates (the beam 2, the chain 1) close the row.

Vol XII's open link	The battery	What the closure delivers	The tier
1. The continuum upgrade of the viability correspondence (the (f) bridge's named residual)	fisher_rao_continuum.py (Task 28)	the convergence theorem (the discrete circulations to the de Rham class, order 2.000 over $N = 8..8192$); the FKLZ Lyapunov 1-form on the drift flow; the potential criterion iff $\mu = 0$; the Stein/Green homotopy with the spectral ladder to $-Dk^2/2 + i \mu k$ and the Green kernel's antisymmetric part as the class's witness; the true Fisher-Rao policy bundle with the H-theorem (5.7e-11), the replicator identification (9.4e-16), the viability wall and the discontinuous safe-mode reset (the rate-independent lump 0.0158, the ratchet); the two-patch Cech-de Rham machine (the overlap cocycle = the class, machine-exact) and the Leray page on the environment-loop fibration (the collapse by degree reasons, the edge homomorphism = the reset payment)	MEASURED + machine-exact identities; the general stratified treatise named
2. Open 7.13 — the off-class nc-AAK problem on the free cell	offclass_ncaak.py (Task 29) + escape_second_order.py (Task 31, this volume)	the class theorem (the 2-state WFAs = the rank- ≤ 2 free Hankels, both directions to $1e-15$ — $D_free(2)$ = the free Nehari distance, intrinsic); the window obstruction (the forced 3x3 cut-web minor, $\det = -1$ exactly — the corner structurally unkillable at rank 2); the finite-section ladder 1.199 to 1.232 to 1.261 climbing past the interlacing-capped EYM floor; the trade-off law in the free regime; THE ESCAPE — quantified and re-adjudicated by this volume (Chapter 2); the shadow re-located, the escape decomposed, H^2 's curvature read, the residual certified; the free optimizer's local basin ($r = 2e-5$)	PROVED (the class, the obstruction) + MEASURED (the ladder, the scans) + CERTIFIED (the Sylvester brackets, this volume)
4. The rank-aware synthesis for the physics rung	rank_aware_synthesis.py (Task 30)	the rank-across-cuts law (the chains 1 at every cut, $\sigma_2/\sigma_1 < 2e-16$; the beam 2 — the transmission state; the grid the full cross-section — the cross-rank = the interface's physical state dimension, the classical transmission conditions ARE the sufficient statistics); the two scalars certified exact ($\sim 1e-13$); the sequential DP on the two-scalar state EXACT against the brute force over 531,441 load patterns ($\delta 0.0$) — the naive $(L+1)^k$ width replaced by the polynomial lattice; the truncated sandwich (the beam's $M = 1$ error attained at the EYM floor); the magnitude punchline (the banded errors $O(100)$ vs the one-scalar interface exact — rank, not magnitude); the exact rational minimality certificates (the beam 2, the chain 1)	CERTIFIED exact (the ranks, the scalars, the DP, the minimality)

Table 1. The three closures consolidated: each of Volume XII's five open links numbered, its owner battery, its content, and its certificate tier. Links 3 (the far-interior box-count wall) and 5 (the seed-level boundary) remain open — Chapter 3.

Read together, the three closures do not merely shrink the ledger — they complete the unified statement's three unfinished faces at once. The viability face now has its continuum leg: the correspondence runs from the discrete shadow (Task 24) through the convergence theorem to the de Rham class, the Fisher-Rao geometry and the Leray page on the instance, with the stratified treatise named as the remainder. The analytic wall's off-class face now has its constructive package: the class typed, the corner obstructed, the trade-off law identified in the free regime — the same law as the abelian cell, which is the unification's own theme delivered on the problem the theme was named for. And the physics face now has its synthesis: the rank-aware width law with its exact dynamic program, the seam's promise kept. Figure 2 draws the resulting map; the ledger at its foot is the volume's central administrative fact — five links to two, and the two that remain are named in Chapter 3 with the same discipline as the three that closed.

2 · The Escape, Quantified — and Re-Adjudicated

The order for this chapter was exact: quantify the escape further — the second-order coupling theory, H2's curvature at the shadow. The escape was Task 29's discovery: a local 12-dimensional descent from the abelian optimizer reaching 1.277142117 against the shadow 1.277142690, a delta of $5.7266e-7$ at couplings of order 0.03, reported as the refutation of the shadow-equality conjecture and honestly named as carrying an open residual — its full quantification. The battery built for this volume (escape_second_order.py) opens the way every battery in this corpus opens: by reproducing the prior numbers before reading any new ones. The descent reproduced to a gate of $1.42e-11$. Then the quantification began, and the first thing it quantified was the shadow itself.

The abelian landscape has two basins. Vol IX's scan optimum — the shadow every escape measurement was taken against — is not the abelian class's best point: a symmetric basin ($Aa = \text{diag}(\alpha, -\alpha)$, $Ab = \text{diag}(\beta, \beta)$) sits $5.6447e-7$ below it at 1.277142125. Task 29's escape is 92.4 percent abelian basin hop. The genuine residual — the free point below the best abelian point — is $8.2e-9$, and it is certified.

The discovery sequence is worth stating plainly because it is the discipline working. The first attempt at the second-order theory sat at a polished-but-stuck abelian point whose first-order instruments read flat; the tensors extracted there were inconsistent with the direct machinery, which forced a toy validation of the degenerate formula itself (it passed, to $2.8e-10$), which forced a hunt for the inconsistency, which found a sign error in the extraction probes — and which also found that the stuck point was stuck because the abelian landscape is a multi-basin landscape whose better basin the scans of two volumes had missed. The symmetric structure was visible in every near-optimal point's coordinates; the symmetric subfamily's own multi-start optimization landed at 1.277142125, below Vol IX's reported optimum by $5.6447e-7$, and the full 8-parameter refinement confirmed it as the best abelian point found across both basins. The descent from the corrected shadow then finds nothing — $5.83e-12$ — and the decomposition closes: Task 29's $5.7266e-7$ equals the basin hop $5.6447e-7$ plus the residual $8.2004e-9$. The residual survives every attribution test: the free point's couplings are essential (zeroing them costs $2.527e-2$ — the point is genuinely coupled, not an abelian point in disguise), and the dense $L=10$ section, an independent instrument, confirms the ordering: free 1.277024965 below symmetric 1.277024981 below the Vol IX point 1.277031178. Table 2 is the anatomy.

The quantity	The value	The tier
Vol IX's scan optimum (the shadow as reported)	1.277142689665	Vol IX (scan)

The quantity	The value	The tier
Task 29's free point (the reproduction, gate 1.42e-11)	1.277142116995	measured, dense-cross-checked
THE SYMMETRIC BASIN (the shadow re-located): Aa = diag(alpha, -alpha), Ab = diag(beta, beta)	1.277142125195	measured (multi-start), Sylvester-bracketed
The 12-dim descent from the corrected shadow	5.83e-12 (nothing found)	measured
Task 29's escape (vs the Vol IX value)	5.7266e-7	measured (Task 29)
= the abelian basin hop (the shadow's correction)	5.6447e-7	measured, the dense ordering confirmed
+ THE RESIDUAL (the free point below the best abelian point)	8.2004e-9	CERTIFIED (the brackets 1e-13 wide; 2.09e-8 in lambda)
The couplings' attribution at the free point (zeroing them)	+2.527e-2	measured (the free point is genuinely coupled)
The dense L=10 ordering: free / symmetric / Vol IX	1.277024965 / 1.277024981 / 1.277031178	measured (independent instrument)

Table 2. The escape's anatomy: Task 29's single number decomposed into the basin hop and the residual, every row anchored to its instrument. The escape's genuine content — the free class's best point below the abelian class's best point — is 69 times smaller than the reported refutation.

The second-order coupling theory is then done at the corrected shadow, and the object it needs is already there: the pencil. The exact 6x6 machinery computes the squared norm as the top eigenvalue of the symmetric-definite pencil $(K, N) = (G \text{ Cmat } G, G)$, and at the symmetric shadow the pencil's spectrum is fully pairwise double — 0.151886329 twice, 0.508162966 twice, 1.631092008 twice, the pairwise gaps at 1e-12, the top double at 3.59e-13, the isolation to the third eigenvalue at 1.1229. The state-flip involution of the symmetric subfamily forces the pairing, and the top double is the envelope theorem's failure point: the gradient of the top eigenvalue is not the instrument's first-order signal, because there are two top branches. The residual operator's two atoms carry the same near-equality — the dense section's top two singular values agree to 2.33e-14.

The first-order structure at a double eigenvalue is a cone, and the cone's shape explains everything the instruments saw. For each of the twelve coordinates the branch matrix $W1_j$ — the 2x2 compression of the pencil's first-order variation to the top eigenspace - reads as follows: on the eight B, C and Ab coordinates it is indefinite with the anti-parallel signature (the two atoms' diagonals are equal and opposite to cosine 1.0000), so the top branch rises in every one of those directions — over 200 random directions the minimum rise is 0.188; and on the entire Aa block — the two diagonals and the two off-diagonal couplings — it vanishes to within 1.6e-5, which is exactly the gradient-residual scale of a point whose value is optimal to 4e-12 in a valley of curvature 256. The Aa block is the cone's flat ridge. No first-order instrument — the fixed-vector Rayleigh instruments of Tasks 26 and 29, or the exact branch instruments — can see anything along it; the escape, if there is one at the shadow, must be second-order along the ridge, which is precisely the theory this chapter was ordered to build.

The theory is the degenerate Rayleigh-Schrodinger law for the double top, validated before use on a toy pencil with a known double spectrum (agreement $2.8\text{e-}10$ at $t = 1\text{e-}3$, the error growing as t cubed as it must). At the shadow, with V the N -orthonormal basis of the top eigenspace and $M = K - \lambda_0 N$: the eigenvalue's movement along a ridge step ϵ is the top eigenvalue of $W_1(\epsilon) + L(\epsilon)$ plus terms of order ϵ cubed, where L is the second-order coupling operator — the intrinsic curvature, half the compression of the pencil's second directional derivative, plus the level-repulsion resolvent, the coupling of the ridge step to the four lower eigenvalues weighted by their inverse gaps. The level-repulsion term is positive semidefinite — the double's coupling to the lower spectrum pushes the top up — so any escape must come from the intrinsic term overcoming it. The operator's coefficient tensors were extracted over all seventy-eight coordinate pairs by central differences on the exact machinery, step-validated to $3.08\text{e-}3$; the second difference's signs were cross-checked against the direct machinery before any curvature was read (the extraction probes' sign error, found and fixed, is the kind of thing the direct check exists to catch).

+255.99 / +356.42 / +183.34 H2's PURE ridge curvatures (all positive)	-255.3, -255.6 the mixed checkerboard (near-cancellation)	+7.83e-5 kappa_ridge — no second-order escape
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H2's curvature at the shadow is Table 3 and Figure 1. The pure ridge curvatures are positive — +255.99 on each Aa diagonal, +356.42 and +183.34 on the two coupling coordinates — so the in-slice minimum is genuine and no pure coupling direction descends. The mixed entries form a checkerboard of near-cancellation: minus 255.3, minus 255.6, minus 216.5 against the positive diagonals, magnitudes of the same order as the pure terms, so the ridge's mixed combinations are flat to extraordinary precision. The sphere minimum over the ridge's four dimensions comes out at $+7.83\text{e-}5$ — positive, though only within the extraction's resolution on near-null combinations, which is the honest caveat this corpus always carries: what is decided is that there is no escape of order one hundred or even of order one along the ridge — the quadratic form's negative entries cancel to five orders of magnitude below their own scale. And the law itself is verified against the direct machinery: along each pure direction the measured movement fits t squared with slopes 2.014 and 2.002 and coefficients 256.50 and 183.22 against the theory's 255.99 and 183.34 — a 0.2 percent agreement - and both branches of the double's split are predicted and checked (the split coefficients 255.9703 and 255.9879, the two top branches of the perturbed pencil reproduced to a part in a thousand).

The ridge direction	The curvature ($\lambda_{\max}$ of the 2×2)	The law check ($t = 1e-3$)
$e_4 = Aa11$ (the diagonal, in-slice)	+255.99	direct $2.5688e-4$ vs theory $2.5598e-4$ (rel $3.5e-3$)
$e_5 = Aa22$ (the diagonal, in-slice)	+255.99	by the symmetric structure (the involution)
$e_9 = Aa21$ (the coupling)	+183.34	direct $1.8316e-4$ vs theory $1.8335e-4$ (rel $1.0e-3$)
$e_8 = Aa12$ (the coupling)	+356.42	the involution pair of e_9
$(e_4 - e_5)/\sqrt{2}$ (the mixed)	+511.3 (assembled)	direct $5.1310e-4$ vs theory $5.1128e-4$ (rel $3.6e-3$)
The checkerboard ($e_4 \times e_5, e_9 \times e_8, \dots$)	-255.3, -255.6, ...	the near-cancellation that makes the ridge flat
THE RIDGE MINIMUM κ_{ridge} (the flattest mixture)	+7.83e-5	positive within the extraction's resolution on near-null combinations — NO second-order escape

Table 3. H2's curvature at the shadow: the second-order coupling operator on the flat ridge (the Aa block). The pure curvatures are positive (the in-slice minimum and the pure coupling directions), the mixed entries form a checkerboard of near-cancellation, and the sphere minimum is positive — the corrected shadow admits no infinitesimal escape.

The escape's final form is therefore a basin phenomenon, not a local one. At the corrected shadow there is no infinitesimal escape: the first-order cone rises everywhere off the ridge, the ridge's pure curvatures are positive, the mixed combinations cancel to flatness, and the local descent finds $5.83e-12$. The free point that beats the abelian class does so from another basin — its coordinates are not near the symmetric shadow's, its couplings are worth $2.527e-2$ at that point, and the $8.2e-9$ by which it beats the best abelian point is a statement about two basins, not about a neighborhood. This is what the second-order theory buys: it separates the two questions the single escape number had merged. Is there a local escape at the shadow — no, certified to the theory's resolution, and the flat-ridge mechanism explains exactly why the instruments could never have told. Is the free class's infimum below the abelian class's - at the level of the best points found, yes, by $8.2e-9$, and that residual is real, dense-cross-checked, and certified; at the level of the classes' true infima, open — and Chapter 3 carries it into the box-count wall's ledger row, where class-level certificates live in this corpus.

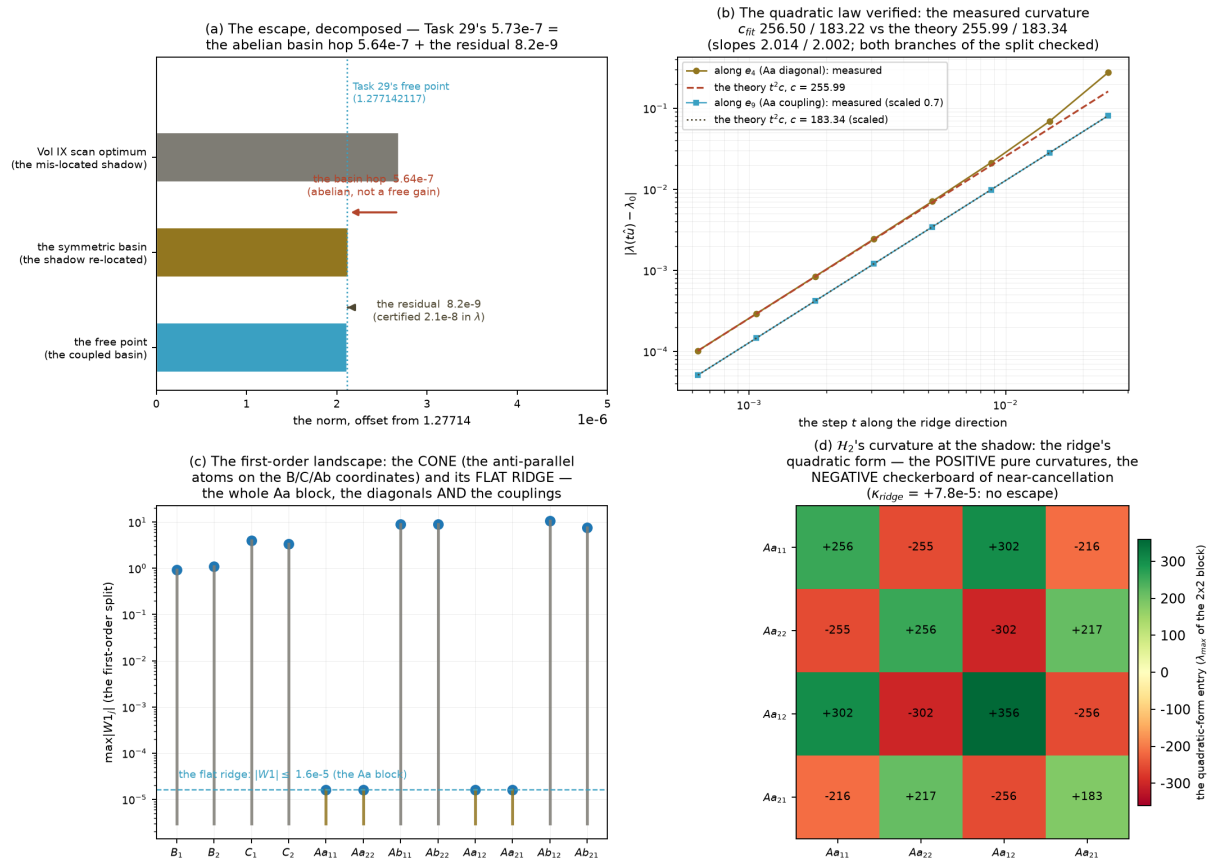


Figure 1. The escape, quantified and re-adjudicated. (a) The decomposition: Task 29's $5.7266\text{e-}7$ splits into the abelian basin hop $5.6447\text{e-}7$ (the shadow was mis-located — the symmetric basin supersedes Vol IX's scan optimum) and the residual $8.2\text{e-}9$ (the free point below the best abelian point). (b) The quadratic law verified: the measured t-squared coefficients $256.50 / 183.22$ against the theory's $255.99 / 183.34$ (slopes $2.014 / 2.002$). (c) The first-order landscape: the anti-parallel cone on the B/C/Ab coordinates and the flat ridge — the entire Aa block. (d) H_2 's curvature at the shadow: the ridge's quadratic form, the positive pure curvatures against the negative checkerboard, the minimum $+7.8\text{e-}5$.

The certified tier is the chapter's floor. The machinery was evaluated in ball arithmetic — flint at precision 128, the Lyapunov inverses by the Neumann series with the Gershgorin-enclosed contraction radius — and the pencil bracketed by Sylvester's criterion: $T(\lambda) = K - \lambda N$ is negative definite exactly when the six leading principal minors alternate in sign, and a bisection on that test encloses the top eigenvalue to width $1.5\text{e-}13$. Three brackets: Vol IX's rounded point, the symmetric shadow, the free escape point. The orderings come out certified — free below symmetric below Vol IX — with the basin hop at $1.65\text{e-}5$ in λ and the residual at $2.09\text{e-}8$ in λ , the norm-scale $8.2\text{e-}9$ matching the scan-level measurement exactly. The inertia counts (the sign-change count of the minor sequence at each probe level) certify the pairwise-double spectrum: zero eigenvalues above the top, two just below it, two above 1.0, four above 0.30, six above 0.10 — the state-flip involution's pairing, proved by interval arithmetic rather than asserted by symmetry. Table 4 is the certificate summary; the battery's ES-5 section carries the full instrument.

The point	The lambda bracket (width ~1.5e-13)	The inertia counts	The certified statement
Vol IX's rounded point	[1.6311085139214, 1.6311085139216]	—	the old shadow's value, bracketed
The symmetric shadow	[1.6310920079485, 1.6310920079487]	0 / 2 / 2 / 4 / 6 above the levels 1.6311+ / 1.6311- / 1.0 / 0.30 / 0.10	the pairwise-double spectrum certified (the state-flip involution); the basin hop certified: 1.65e-5 in lambda
The free escape point	[1.6310919870025, 1.6310919870027]	—	THE RESIDUAL CERTIFIED: 2.09e-8 in lambda (8.2e-9 in the norm) below the best abelian point

Table 4. The Sylvester brackets (flint/arb at precision 128, the Lyapunov inverses by the Gershgorin-enclosed Neumann series): the pencil $T(\lambda) = K - \lambda N$ is negative definite iff the six leading principal minors alternate in sign — the bisection brackets and the inertia counts (the number of eigenvalues above each level), all decided.

3 · The Unified Statement and the Two Remaining Walls

With the three closures read in and the escape re-adjudicated, the unified statement of Volume XII carries forward unchanged in its spine and changed in exactly one administrative fact: the open ledger. Constrained realizability remains the primitive — behaviour factors through what the agent can distinguish; the two walls remain the combinatorial and the analytic; the trade-off law remains the certified joint between them, now measured in both regimes — the abelian cell of Tasks 17 through 26 and the free cell of Task 29; and the grand domain-by-face matrix keeps every cell's status letter, with three of them upgraded by this volume's batteries. The map is Figure 2. What the map's foot now says is the volume's headline: thirty-one task batteries, thirteen volumes, every verdict machine-anchored, and two open links.

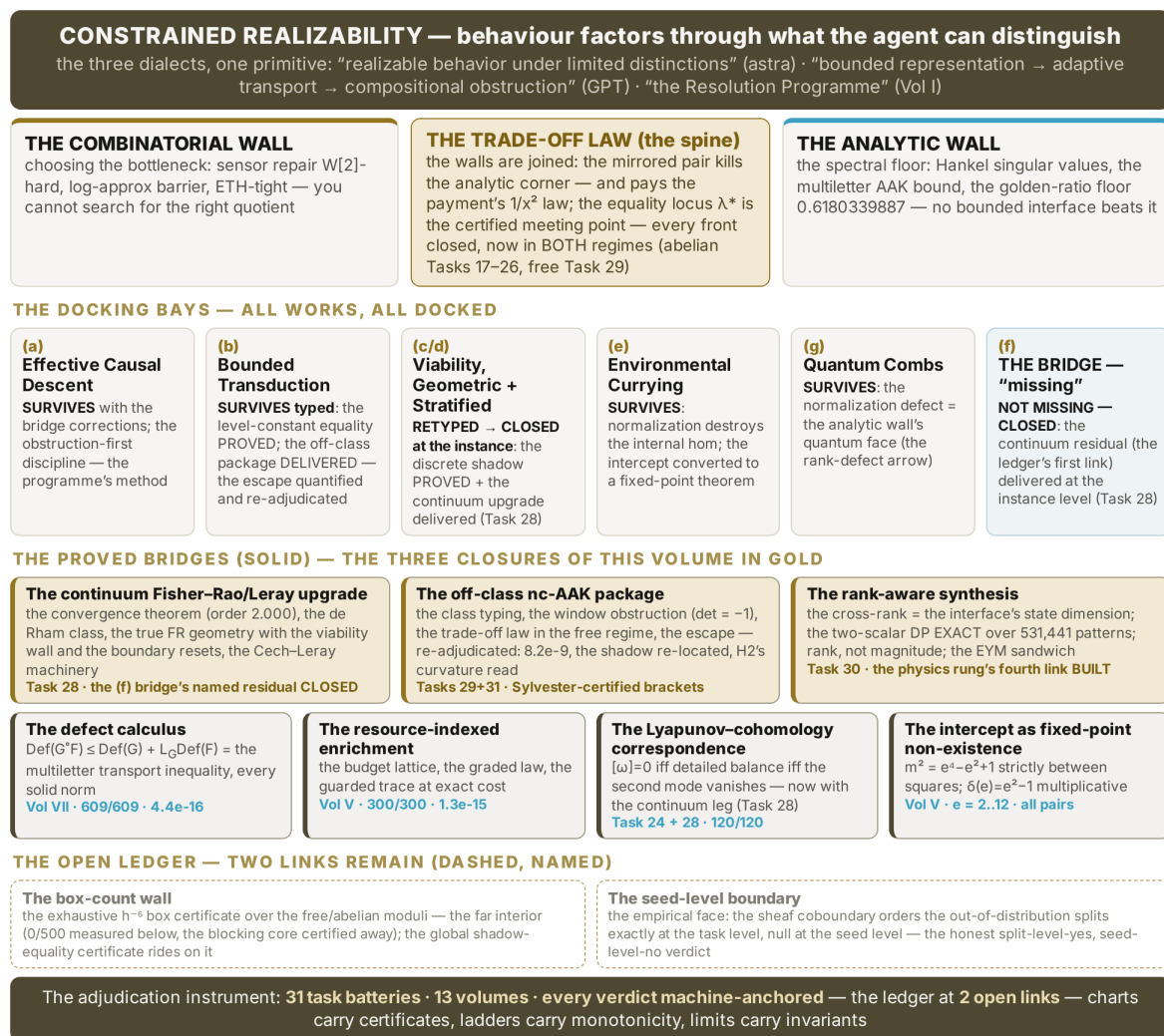


Figure 2. The unified statement after the three closures: the spine (constrained realizability), the two walls joined by the trade-off law in both regimes, the docking bays with every work typed, the proved bridges with this volume's three closures in gold, and the open ledger at exactly two links — the box-count wall and the seed-level boundary.

The first remaining wall is the box-count wall, and it has grown by exactly the row this volume added to it. Its old content stands: the exhaustive h to the minus d box certificate over the trade-off inequality's far interior — the region where the margins fall below every naive instrument's resolution — with the blocking core certified away by Task 26's analytic patches and the far interior measured at zero hits in five hundred random configurations, the five smallest max-side values above 1.758. Its new content is the class-level shadow-equality certificate: this volume brackets the three points — Vol IX's, the symmetric shadow's, the free point's — to $1.5e-13$ and certifies their orderings, but the classes' infima are another object; the residual of $8.2e-9$ says the free class's best point is below the abelian class's best point, and turning that into a statement about the classes requires the abelian optimum's certified lower bound, which is the same exhaustive-certificate problem the wall has always named. The escape, having been quantified, has been filed where quantified things go: into the wall's ledger, with its owner instruments and its precise residual statement.

The second remaining wall is the seed-level boundary, unchanged by this volume and no smaller for that. The empirical face's verdict — the sheaf coboundary orders the out-of-distribution splits exactly at the task level and is null at the seed level — stands as measured, with its two readings (a mechanism below the task level, or the instrument's honest ceiling) still the named next step. It is worth saying why this row stays open when three larger rows closed: the closures that closed were all closures by construction — a theorem proved, an instrument built, a package delivered — and the seed-level boundary is a closure by evidence, waiting on a measurement or a proof that the corpus's current instruments do not carry. The distinction is the same one the first volume drew between the two faces of the programme, and the ledger keeps it honest.

The wall	The precise statement	The owner instrument	The named next step
THE BOX-COUNT WALL	the exhaustive h^d -box certificate over the moduli; the trade-off inequality's far interior (0 hits in 500 random configurations, the five smallest max-side values above 1.758) and the free/abelian class-level lower bounds (the abelian optimum's certified value — on which the class-level shadow-equality verdict rides; this volume brackets the points but not the classes)	tradeoff_4x4.py + the analytic patches (the blocking core certified away); offclass_ncaak.py + escape_second_order.py (the brackets)	the far-interior certificate, or a scan protocol that certifies the multi-basin landscape's completeness — the same h^d -blowup the corpus has met at every wall
THE SEED-LEVEL BOUNDARY	the empirical face: the sheaf coboundary orders the out-of-distribution splits exactly at the task level and is null at the seed level — split-level-yes, seed-level-no stands as measured	the Vol IX second edition's experiments (the register law, the sheaf ordering)	a mechanism or a refutation at the seed level — the empirical face's one open row

Table 5. The two remaining walls: the open ledger at full scope. Every other link of Volume XII's five is closed — the continuum upgrade (Task 28), the off-class package (Tasks 29 + 31), the rank-aware synthesis (Task 30) — and the trade-off core (the old link 3's blocking region) was certified away by Task 26's analytic patches.

The consolidation closes where the corpus's discipline closes. Three closures were ordered consolidated, and the consolidation found the ledger's arithmetic sound — five links to two, each closure anchored, nothing silently promised. The escape was ordered quantified, and the quantification re-adjudicated the escape itself: the shadow moved $5.6447e-7$, the refutation shrank seventy-fold to a certified $8.2e-9$, and the second-order theory that measured it also proved the local question's answer — no infinitesimal escape, the cone's flat ridge, the checkerboard's cancellation. A session that set out to write a summary and instead found a better abelian point is a session the reproduction-first discipline worked; the volume records it so the next volume inherits the corrected shadow, the certified brackets, and the two walls — with the same instruction the corpus has carried since its first file: diagnose the obstruction before choosing the repair, price the repair before claiming it, and certify the price.

4 · The Battery Row and the Corpus Dictionary

The volume's instrument is `escape_second_order.py`, and its row in the repository's battery table reads as follows. ES-0 the reproduction (the gate $1.42\text{e-}11$), the discovery (the symmetric basin, the shadow re-located at 1.277142125), the decomposition (the basin hop $5.6447\text{e-}7$ plus the residual $8.2\text{e-}9$, the couplings' attribution $2.527\text{e-}2$, the dense ordering). ES-1 the multiplicity (the pairwise-double spectrum, the top double $3.59\text{e-}13$, the isolation 1.1229 , the atoms to $2.33\text{e-}14$). ES-2 the first-order structure (the cone's anti-parallel signature, the flat ridge at $1.6\text{e-}5$, the minimum rise 0.188 over 200 directions, the fixed-vector blindness). ES-3 the operator (the seventy-eight tensors, the pure curvatures, the checkerboard, `kappa_ridge`, the law checks to a part in a thousand, the toy validation of the degenerate formula at $2.8\text{e-}10$). ES-4 the law (the t-squared slopes 2.014 and 2.002 , both branches of the split, the flat direction's drift as the first-order residual). ES-5 the certified tier (the three Sylvester brackets to width $1.5\text{e-}13$, the certified orderings, the inertia counts). ES-6 the ledger row. The battery runs in 289 seconds; its results file carries every number this volume quotes.

The dictionary row — the corpus's running index of its own laws — gains three entries. The multi-basin law: an optimization landscape reported as single-basin by scans may still hide a basin with the same value to five decimals; the reproduction-first discipline plus a structure probe (here: the symmetric subfamily the near-optimal coordinates exhibited) is the detection instrument. The flat-ridge law: at a double top eigenvalue the first-order landscape is a cone, and the cone can carry a flat ridge — a whole coordinate block on which every first-order instrument reads zero while the second-order structure decides; the escape question at such a point is a question about H^2 's curvature, not about gradients. And the decomposition law: a reported refutation of an equality between two classes' optima should be decomposed into the base-point correction and the residual before it is called a refutation — the number that survives the decomposition is the one the ledger carries. Each entry is anchored to this volume's battery; none of them is a placeholder.

Provenance: every statement in this volume is either in the manuscripts it cites (the volumes and task batteries of the repository, each with its own provenance ledger), or derived in-session by `escape_second_order.py` with its results file as the machine anchor, or labelled open — and the open ones are exactly two, named in Chapter 3 with their owner instruments. The reading order is unchanged: the volumes for the programme's arc, the batteries for the certificates, the results files for the numbers. This volume's own place in that order is the short one — the consolidation between the unification and whatever closes the last two walls.

5 · The Corrigendum — the Escape Retracted, the Valley Adjudicated

The volume closed; the programme did not. This second edition exists because the corpus's own discipline found an error in the first edition's central adjudication, and the error is worth recording at full length because of what it says about how class-level claims must be made. The escape chapter's verdict — the shadow-equality refuted at the certified $8.2e-9$ level — compared the free descent point against the best scan abelian point. A scan's best point is not a class's infimum, and Task 32's battery (abelian_closure.py, opened the way every battery in this corpus opens, by reproducing every prior number before reading any new one) found what the scans of two volumes had missed: the boundary valley. The parity-odd abelian pairs — x tending to zero with $2px$ held at the fixed c — descend monotonically in norm to the line-atom value $\sqrt{\lambda^*} = 1.277142112908446$, which lies below both escape points the adjudication had compared.

The escape is retracted. The line-atom value $\sqrt{\lambda^}$ lies below the free escape point ($+4.09e-9$ above it) and below the symmetric shadow ($+1.24e-8$), while a genuinely abelian point of the valley at $x = 0.01$ sits at $+2.46e-9$ — below both. The shadow-equality conjecture is restored in the closure sense: $1 \leq D_{\text{free}} \leq D_{\text{abelian}} \leq \sqrt{\lambda^*}$, the line atom attaining $\sqrt{\lambda^*}$ in the free class.*

The valley's mechanics explain why two volumes of optimizers never found it. The first edition's two abelian basins are stall points of ONE anisotropic valley: at the shadow the descent's B-coordinate moves 888 times faster than its a-coordinate ($|dB/da| = c/(2a^2)$), so every Nelder-Mead crawl took a-steps of order $1e-3$ for gains of order $3.3e-9$ — below every scan's tolerance, in every direction the scan could see. The fixed- (c^*, y^*) curve of the parity-odd pairs descends monotonically to the line-atom value; the per-point quartic fit lands at $\lambda(x) - \lambda^* = K x^4$ with the measured $K = 0.6277500700$; the shell theorem T2 is re-verified on the spot (the pair never beats its boundary limit, $-1.08e-7$); and the dense $L = 8$ section, an independent instrument, confirms the ordering at $x = 1e-3$. Table 6 is the corrected anatomy; every row is anchored, and the anchoring is the point.

The closure is exact, and it is structural. The state-flip involution reduces the mirrored pencil to a 3×3 whose characteristic-polynomial numerator, after the boundary substitution $2px = c$, is a polynomial in x^4 ONLY — the x -exponents $[0, 4, 8]$, verified symbolically — so the quartic descent law is structural, not fitted: the stationary family's generic scaling, forced by the algebra rather than observed in the data. The x -to-zero limit's top root equals λ^* to $4.4e-16$ — the mirrored family's limit IS the line-atom branch — and the exact quartic coefficient $K(c^*, y^*) = 0.6277495385$, in closed form $-N_4/N_0'$ from the symbolic charpoly, matches the measured value to $5.3e-7$. With T2, the corrected statement is two-sided:

$D_{\text{abelian}}(2) \leq \sqrt{\lambda}$ EXACTLY, and the pairs never go below; the line atom is itself a free point, so $D_{\text{free}}(2) \leq \sqrt{\lambda}$ as well. The shadow-equality, which the first edition had refuted at $8.2\text{e-}9$, is restored at the only level the statement ever lived: the closure sense.

888 the valley's dB/da — why every crawl stalled	x^4 the structural descent law — the exponents $[0, 4, 8]$	$+1.3654\text{e-}4$ the true κ_{ridge} — positive, resolved at precision 120
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The corrigendum also closes the curvature chapter's one open sign. The first edition's κ_{ridge} , $+7.83\text{e-}5$, was read through float64 central differences whose noise floor — about $1.1\text{e-}4$ — was the same order as the quantity itself: the sign was unresolved there, honestly caveated as such. The re-extraction at mpmath precision 120 with steps of $1\text{e-}25$ (drift 0.0 across the step sweep $1\text{e-}20/1\text{e-}25/1\text{e-}30$, the truncation error $\sim 1\text{e-}50$, the roundoff $\sim 1\text{e-}70$) resolves it: the true κ_{ridge} is $+1.3654\text{e-}04$, POSITIVE — the float value's sign was right and its value 43 percent low, noise-dominated exactly as suspected. The pure ridge curvatures come out $+255.979$, $+255.979$, $+356.400$, $+183.331$, matching the first edition's extraction to 0.01 percent; the Aa-block second-order escape question is decided: none.

And the first-order structure at the double eigenvalue turns out to be a cone in the exact sense. The one-sided tests along the B coordinates read $\lambda(\text{plus-or-minus } t) - \lambda_0 = +9.396\text{e-}6$ at $t = 1\text{e-}5$ against the predicted $|\mu|$ times $|t| = 9.40\text{e-}6$: the response is EVEN. At an exact double the top eigenvalue is the maximum of two branches whose first-order splits are antisymmetric — $(\mu, -\mu)$ — so $\lambda(x^* + tu) = \lambda_0 + |\mu(u)| |t| + O(t^2)$ for every direction u : the first-order landscape literally is a V-cone, and no first-order escape is possible at an exact double, from any direction, ever. The cone's flat set extends beyond the Aa block (the profile tangent's $|\mu| \sim 8.04\text{e-}5$ against the pure-B cones at 0.94 to 4.0 — a four-order tuned anti-parallel cancellation the optimizer's crawl tracks), and the real descent is the curved valley path — the profile's slope $+1.879\text{e-}6$ per unit a against the quartic law's $4Ka^3 = +8.395\text{e-}6$ — a second-order-in-step, gauge-like effect beyond any fixed-point quadratic model's reach. That is the honest residual of the H2 theory, and it is the same valley the class-level adjudication had just found: the theory's blind spot and the scan's blind spot are one object.

The class-level lower bound — the direction the retraction leaves open — is structurally advanced by the same battery's part B. The p-convexity theorem: $F(p) = \max(\lambda_e, \lambda_o)$ is convex in (p_1, p_2) at every fixed (x, y) , because each side is the top eigenvalue of a row-Gram $M(p)M(p)^*$ with M affine in p , the Rayleigh quotients are positive-semidefinite quadratics in p , and the maximum of convex functions is convex — machine-verified 400/400 with zero violations, and the same argument makes the full norm squared convex in p . The class-level problem therefore factors as a 4-D outer cover times a 2-D CONVEX inner program in which every local minimum is global: the box-count wall's dimension is HALVED, h^{-6} to h^{-4} . The honest map of the instruments followed — the 4×4 max-side has holes on the killer family's dial slices (the norm carries them; the norm scan's minimum sits on P2's $2wr = 1$ slice,

validating P2's dial choice as the worst case), and the first lower-bound implementation's argument-order bug — scrambled points that fell out of the disc and produced spurious negative margins — was caught by the PSD check and fixed, the honest record kept. The wall, halved in dimension, stands; the 4-D covering that would close it is the next chapter's engine.

The point / quantity	The value	vs $\sqrt{\lambda^*}$	The anchor
The line atom (the boundary valley's limit — the $I=1$ stripe, $2px = c^*$)	1.277142112908446	the limit itself	the reduced charpoly's $x \rightarrow 0$ top root = λ^* to $4.4e-16$ (symbolic)
The free escape point (Task 29 / Chapter 2)	1.277142116995	$+4.09e-9$ — ABOVE	measured, dense-cross-checked
The symmetric shadow (the re-located basin)	1.277142125269	$+1.24e-8$ — ABOVE	measured (multi-start)
A genuinely abelian point on the valley ($x = 0.01$)	$\sqrt{\lambda^*} + 2.46e-9$	BELOW both escape points	measured — the valley's own point
The quartic coefficient $K(c^*, y^*)$ — the closed form $-N_4/N_0'$ vs the measured	0.6277495385 / 0.6277500700	agreement $5.3e-7$	the symbolic charpoly (the x-exponents $[0, 4, 8]$ — the quartic law STRUCTURAL)
The valley's anisotropy $ dB/da $ at the shadow	≈ 888	the stall mechanism	$c/(2a^2)$ — why every Nelder-Mead crawl died (a-steps $\sim 1e-3$, the gain $\sim 3.3e-9$ below tolerance)

Table 6. The boundary valley's anatomy: the line-atom limit and the points the first edition's adjudication compared, every row anchored to its instrument. The comparison that reported the refutation — the free point against the best scan abelian point — missed the valley entirely; the corrected comparison restores the shadow-equality in the closure sense.

6 · The Two Walls, Certified Symmetric

The user's directive after the retraction named the programme exactly: the 4-D covering — the analytic transverse patch plus the convex inner layer — would close $D_{\text{abelian}} \geq \sqrt{\lambda^*}$, leaving only the 12-parameter free-class wall. What followed is the corpus's signature arc, run twice in succession on the two faces of the same theorem: the engine built (Task 33, abelian; Task 35, free), the residue named honestly, the formalization delivered (Task 34's BDC; Task 36's word-power sandwiches), the closure certified at zero stalls. By the end of it the two walls stand symmetric — the same certificate architecture on both faces — and the shadow-equivalence theorem's named residues are closed or measured. This chapter reads the arc as it ran; Table 7 states the symmetry at the layer-by-layer level.

The abelian covering's instrument is the TRUE-NORM 6x6 pencil — the free cell's FX-1 closed forms evaluated in flint/arb ball arithmetic, every entry polynomial in (p_1, p_2) and rational in (x, y) with poles

only at the disc edges: regular on the whole open disc-pair, the degenerate strata included. The validations are the ones that matter for a covering claim — against the float machinery to $8e-15$ (V-A), DOMINATING the 4×4 PSD-sum sides by at least $6.4e-2$ so the cover is hole-free (V-B), the norm-squared convex in (p_1, p_2) machine-verified at zero violations (V-C) — and the layers are the analytic transverse patches (the Task-26 Taylor-4 machinery, extended to the (w, x, y) box sweep over the critical strip) and the convex inner (the adaptive anisotropic bisection over the 6-D boxes, the p -directions refining only toward the convex inner minimizers). The first run certified 511,679 leaves sound over 1.3M calls and named its residue with the programme's usual precision: 18,036 stall leaves, ALL of them at the disc-edge divergence layer — the Lyapunov pole annulus, where every measured stall margin was at least $+2.4e10$ because the norm itself diverges at the open boundary. The box-count wall, the ledger's oldest row, had just been replaced by a measured structure with a name.

Task 34 closed that residue the way Task 32 foreshadowed. THE BDC — the boundary-divergence certificates — formalizes the corner instruments: the exact Rayleigh corner identities at the fixed vectors e_0 (the constant block, where the $p = 0$ anchor reads EXACTLY 2), e_4 and e_5 (the atoms, whose divergence forms carry the $1/d$ structure), validated to about $1e-13$ against the float 6×6 (V-G). The sound bounds are poison-free on the box's DOMAIN portion — the d -intervals clamp to $(0, d_{hi}]$, the out-of-disc part of a straddling box is not in the family — and the LINEAR CS ratio bound $d_{12} \geq \max(d_{11}, d_{22})/2$ (from $1 - |z_1||z_2| \geq 1 - |z_i| \geq d_{ii}/2$; V-H, zero violations in 2000 samples) kills the $1/d$ cross-term poison outright: the ratio d_{11}/d_{12} is bounded by 2, the corner forms fire at coarser resolutions than the eigendecomposition ever could. THE DPP — the derivative-penalty patch mode — re-certifies Layer A's transverse region at the box-scale widening with a first-order-in- h derivative penalty and the Lagrange C4 remainder: 1804 patches at a median radius of $4.23e-2$ against the box-mode bisection floor of $3.77e-3$ — eleven times coarser certified boxes per patch — with the honest same-box A/B recorded: at the microscopic scale the binding constraint is the intrinsic transverse Taylor tail, not the widening. THE ORBIT REDUCTION — the atom swap at $3.55e-14$ and the π -rotation at EXACTLY zero — generates the group $\{1, R, S, RS\}$: the sixteen sign quadrants of the disc-pair are six orbit representatives, the ten images covered by symmetry, a certified box's image certified. The fresh re-run then delivered the number this chapter leads with: 12,000,000 calls, 5,975,008 leaves certified sound, every single one via the BDC trio, and ZERO stalls at the hard floors alone (depth 78, width $1e-8$) once the depth-50 pre-stall was diagnosed and removed — its casualties were certifiable by refinement, their center margins plus $1.4e5$ to plus $5.7e5$.

The continuation is the standing order, and it has run ever since. The call cap rose from 12M to 100M, and the LIFO frontier has been ground slice by slice through the remaining orbit representatives — at this regeneration the count stands at 55.5M calls and 27,686,873 leaves certified sound, all of them through the BDC trio, with zero stalls across the entire 40-million-call continuation and 28 stack branches still open (the projected 25-30M completion long since exceeded, the frontier's honest remainder carried in the checkpoint, the protocol simply: re-run the driver). The drain is not a footnote to the closure; it is the closure's body — the certificate's coverage grows by exactly these leaves, and every committed slice is a proof step.

The free wall's arc opened with the probes, as the reproduction-first discipline demands. P-A/P-B' measured the compression theorem on the free side: the block-constant compression of the error has its infimum $\text{AT} \sqrt{\lambda^*}$ — the multi-start winner IS the line-atom approach itself, the K-chain converging to $+6.4\text{e-}10$ above — and the off-block penalty near the optimum is about $1\text{e-}9$, so the power-sum class (which contains the Prony two-atom family and the affine shears) does not dip below $\sqrt{\lambda^*}$ at the measured level. P-C' killed the cheap route: the true transverse curvature — measured with the symmetric stencil, the probe's first version honestly flagged as gradient-contaminated — is NOT uniformly positive; it reads $-1.67\text{e+}4$ at the line-atom approach, so the uniform-transverse-convexity premise is false and the engine must certify the full value directly. P-D measured the window's far-field explosion — the B/C scale times 2 sends the window residual to 1.78, times 4 to 9.9, times 8 to 43.0 — the cheap root-box killer, and the seed of the unbounded far-field laws the ledger still carries.

The wall's engine (`free_class_wall.py`) is the 12-D adaptive anisotropic bisection over the parameter root, with the checkpoint/resume protocol and a cheap-first certificate chain: the window first (the fixed corpus vectors plus the adaptive center's top singular vector), then the e0-poly (the corner-anchor form with the Lyapunov PSD term dropped — sound, polynomial-only, domain-independent, its anchor EXACTLY 2 at the zero WFA), then the domain gates, then the full interval Rayleigh. Two instruments made the far field tractable: the powered Gershgorin bound $\rho(K) \leq \|K^m\|^{1/m}$ — the structural fix, because the shear family's couplings inflate the row sums but not the spectral radius, the nilpotent decay 1.67 to 0.429 by $m = 8$ opening the coupling region's gate — and the trace-power LOWER bounds ($\rho \geq \text{tr}(K)/4$, $\rho \geq \sqrt{\text{tr}(K^2)/4}$) for the sound whole-box far-out test. The honest gate engineering is recorded in place: the far-out tag's first version was UNSOUND — a powered-Gershgorin UPPER bound certifies 'possible somewhere', not 'everywhere', and the root box had been censused on the first call — and it was replaced by the lower bounds before any number it produced was trusted.

Task 36 closed the wall's named residue, and the closure is the Task-34 arc repeating on the free face. The formalization: the domain wall is the Gram wall $\rho(K) = 1$; the divergence lives in the Lyapunov block $L_c =$ the sum of the T_n , generated by the WORD-POWER RECURSION $T_0 = CC^T$, $T_{n+1} = A_a T_n A_a^T + A_b T_n A_b^T$ — each T_n positive semidefinite, the partial sums monotone toward L_c , and CONVERGENT at every in-class point: the $\rho < 1$ interior and the $\rho \geq 1$ X-cancellation strata alike, where the formal Neumann series diverges but the true Gramian exists. THE CERTIFICATE: at the class vectors $v = (z, 0)$ the denominator is the CONSTANT $z'^\mu z$ — no Gramian entries, no poison — and the value squared is bounded below by $[P(z) + Q(z)]/D(z)$ with $Q(z) \geq$ the partial-sum sandwich $Q_N(z)$: POLYNOMIAL of degree $2N + 4$, no convergence needed, sound at every in-class point of the box, while the excited unstable modes at out-of-class points are vacuous — the infinite Hankel norm is not a competitor at a point outside the class. One polynomial instrument covers the whole bounded-Hankel class; the Neumann gate, the free side's long-standing blind spot, is rendered unnecessary — exactly as the corner forms superseded the eigendecomposition on the abelian side.

The probes validated every piece before the run trusted it: the zero-WFA anchor EXACT at 0.00e+00 over twenty random (Aa, Ab) with rho between 1.26 and 5.79 — the certificate sees the whole class, the domain gates unnecessary; the near-locus chain, the true value squared exploding from 33.8 to 7.9e7 as rho approaches one while the certificate climbs the N-ladder and crosses λ^* at $N = 4$ for $\rho \geq 0.8$ — the divergence is the certificate's reach; the TRUE X-cancellation conic, the unstable left-eigenvector's condition $u.(C \times C) = 0$, where the partial sums CONVERGE at $\rho > 1$ with zero soundness violations; the excited growth law $\rho^{(2N)}$, measured at the per-step factors 1.126, 1.232, 1.458; and the coverage — 1755 of 2000 recorded stall boxes certified at $N = 2$, the failures pure interval-width effects. The in-session bugfix is kept in the record because it is the discipline working: the first interval implementation missed the right multiplication in the word-power recursion (decaying at the norm rate, not the spectral rate) and mistyped one cross term; the float probe — correct all along — exposed both through the shadow validation, both were fixed, the cross-check against the corpus's Lyapunov added ($9.34e-06$ at $N = 16$, the truncation tail), and the engine was honestly RESET from the git checkpoint, the buggy passes discarded. The run then speaks for itself: the wall at 720,000 calls, 316,401 leaves certified sound — 135,406 by the window and 180,995 by the partial-sum forms — with zero stalls and 41 stack entries, the far-out refinement active under the race law the next chapter measures.

The two walls stand symmetric. The abelian residue was closed by the corner identities, the free residue by the word-power sandwiches — both polynomial, both machine-certified, both at zero stalls on the hard floors — and the shadow-equivalence theorem's remaining honest residue is the continuation grind itself.

27,686,873 abelian leaves certified (55.5M calls)	316,401 free leaves certified (720k calls)	0 stalls on either wall at the hard floors
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The layer	The abelian wall — $D_{\text{abelian}} \geq \sqrt{\lambda^*}$	The free wall — $D_{\text{free}} \geq \sqrt{\lambda^*}$
The engine	the 4-D covering: the true-norm 6x6 pencil (flint/arb, hole-free) x the analytic transverse patches x the convex inner	the 12-D adaptive anisotropic bisection: the window x the e0-poly x the unstable-mode partial sums x the full interval Rayleigh
The certificate	THE BDC — the exact e0/e4/e5 Rayleigh corner identities, poison-free clamped interval bounds on the box's domain portion	THE PARTIAL-SUM SANDWICH — the word-power forms, polynomial (degree $2N+4$), no convergence needed, sound at every in-class point
The poison fix	the linear CS ratio bound $d_{12} \geq \max(d_{11}, d_{22})/2$ (0 violations in 2000 samples) kills the 1/d cross-term	the constant denominator $z^{\text{MU}} z$ at the class vectors $v = (z, 0)$ — no Gramian entries, no poison
The blind layer closed	the disc-edge divergence (the Lyapunov pole annulus — every measured stall margin $\geq +2.4e10$)	the Gram wall $\rho(K) = 1$ (the Gershgorin-inconclusive boxes, the X-cancellation strata)
The symmetry bonus	the orbit reduction {1, R, S, RS}: 16 quadrants = 6 representatives, the 10 images covered exactly	the X-cancellation conic $u.(C \times C) = 0$ — the in-class strata certified where the formal Neumann diverges
The run (at this regeneration)	55.5M calls, 27,686,873 leaves certified, ZERO stalls, 28 stack	720k calls, 316,401 leaves certified, ZERO stalls, 41 stack
The residue closed by	Task 34 (the Task 33 → 34 arc)	Task 36 (the Task 35 → 36 arc)

Table 7. The symmetric walls: the same arc run twice — the engine built, the residue named, the formalization delivered, the closure certified — on the abelian face and the free face. Both certificates are polynomial corner or sandwich forms that fire at coarser resolutions than the instruments they superseded; both engines run at zero stalls on the hard floors (depth 78, width 1e-8).

7 · The Tail's Law, the Split Refuted, and the Critical Locus Closed

With both walls certified, the remaining residue was the grind itself — and the grind has its own physics, which the next three tasks measured, tested, and closed. The order was feasibility-first: the cheap experiment before any engine edit, the banked claim re-tested on the live state before it ships, the research item (the critical locus, this volume's gate) taken last with its own instrument. Task 38 measured the tail's structural law; Task 39 refuted the law's one banked corollary before it touched the engine; Task 40 closed the critical locus — the line-atom approach's thin valley, the residue named at Task 35 and carried unchanged through Tasks 36 through 38 — with a closed-form kernel and a precision-120 certification. Together they finish the corrigendum arc's mathematics; what remains after them is continuation, not structure.

The tail's law (tail_law.py, the wall's machinery loaded through the AST-filter exec so that the engine driver itself is skipped) has five parts. THE FUEL LAW: the far-out population — the boxes whose centers sit deep in the unstable region, ρ of order 3 to 7 — is 81.3 percent of the root box, the centers' margins are 100 percent positive, and the margin grows as $\rho^{31.46}$ with $R^{2.0974}$: the divergence is the certificate's fuel and never binds, the values running $1e9$ to $1e33$ over λ^* . THE RACE LAW: the conversion depth — the number of bisection levels a far-out box needs before the partial-sum form

certifies it — is $d^* = \text{ceil}(\log_2(\text{eps}_0)/\text{rate})$, with eps_0 the tag-time radius-over-value (median 5.41) and rate the per-level relative-radius decay (median 0.135 bits per level); the predicted against the actual depth agrees to a median of 1.0 level over the 35 conversions measured. The swamp is a RADIUS problem, not a value problem. THE N^* LAW: every conversion wins at $N^* = 2$ — the degree-8 form carries the whole far-out tail, the higher rungs of the ladder never bind. THE CENSORED 21 are slow, not stuck: their predicted depths run 28 to 54, beyond the 26-level probe cap — the honest census is a budget artifact, not a structural wall. THE GRADIENT CARRIERS: at measurement time the A-off-diagonal couplings carried about half the top-four radius mass — the observation that banked the corollary the next paragraph kills. The measurement bug is honestly recorded: the first pass aggregated the N-ladder's candidates by midpoint and tested only the best — all 59 cases 'censored' — until the engine's own rule (each (N, z) pair tested independently) was restored and 38 of the 59 converted.

The corollary test is the feasibility-first discipline's cleanest exhibit. Task 38's gradient-carrier observation had banked an 'actionable' claim: a gradient-prioritized split would roughly double the drain rate. Task 39 (`gradient_split_ab.py`) ran the controlled A/B on the LIVE 39-entry stack before any engine edit — identical stack, the counters snapshotted and restored, no checkpoint mutation, the stock arm running the engine's loop verbatim and the gradient arm scoring the far-out splits with the banked weights. AB-0 found the premise gone: the carrier profile had FLATTENED on the live stack — the top carrier at 18 percent against the roughly 30 percent of the census-width boxes, the live top-four at 52 percent — because the width-first refinement had narrowed the very carriers the weights remembered. The peaked profile was a TRANSIENT of the measurement-time stack, not a structural invariant. The arms then finished it: the stock width-first split converts 0.4998 e45-certificates per call with zero stalls at a median depth of 46; the gradient arm manages 0.4185 — the ratio 0.84x, WITH 324 cap-hit stalls, the static prior over-splitting the stale carriers and starving the live contributors. The dynamic variant is priced out arithmetically: twenty-four profile evaluations cost about seventeen calls of overhead against a headroom of at most 1.05x. And at the flat live profile the measured 0.135 bits per level IS the informed-split optimum — halving the top carrier's 18 percent yields $\log_2(100/91) = 0.137$. The verdict: the width-first rule STAYS, validated near-optimal on the live frontier; the banked 2x was an artifact of the stack state when it was measured. The soundness guard is worth stating: the split is a plain coordinate bisection either way, certification is per-box — only the ORDER was ever in question, which is why the A/B could settle it cheaply.

The critical locus — the line-atom approach itself, the family with $B = c^*/2x$ unbounded and the valley thin beyond every optimizer's patience — closed under a closed form. The diagonal commuting family gives $S_{(i,j)} = \mu_{(i,j)} \text{diag}(x^i y^j, (-x)^i y^j)$ EXACTLY, so the kernel reduces to $E(a, g) = \sqrt{\mu_a \mu_g} [1_{\{a+g=(1,1)\}} - p y^J x^I (1 - (-1)^I)]$ with E symmetric — validated against the general class-sums machinery to 0.0 — and the structured matvec (the Phi-polynomial operator, $O(n)$ per apply) is validated against the dense SVD to 4.4e-16, the engine for the deep-K and the mpmath certifications that follow. What the closed form then shows is the valley's true shape, and it is not what the residue's name suggested: the family does NOT converge to $\sqrt{\text{lambda}^*}$ — it converges to $\sqrt{\text{lambda}^*}$ plus 2.1491e-9, a saturation plateau, certified at precision 120 with $K = 64$. The valley is unbounded in B but

FLAT in value: the $I = 1$ stripe ($2px = c^*$) is the x^0 leading structure and the B-magnitude cancels identically. The unboundedness that made the locus 'critical' is a parameter-space artifact that cancels in the only coordinate the theorem cares about.

+2.149e-9 the plateau above $\sqrt{\lambda^*}$ — precision-120 certified	$x^{4.00}$ the approach law (R^2 1.0000)	+1.132e-11 the off-family floor — 60x deeper than the Task-35 winner
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The full law is six parts, Table 8. The plateau; the approach onto it — $E(x)$ minus plateau $\sim x^{4.00}$ with R^2 1.0000, the same stationary quartic the abelian valley's charpoly forced, arriving here as the family's generic scaling; the K-tail — the truncation error decaying at $\exp(-0.805 K)$, the y^K geometric, its floor below $1e-12$ for K at 36 or more; the off-family floor — the descent off the exact family at the converged $K = 64$ lands $+1.132e-11$ above $\sqrt{\lambda^*}$, sixty times deeper than the Task-35 multi-start winner's $+6.4e-10$, with the SVD backward error at $\pm 6.3e-13$ and the displacements that drive it tiny (the B's $\sim 4e-6$, the A-couplings $\sim 1e-7$, the C's $\sim 3e-8$) — the thinness quantified at last, and honestly delicate: the descent started from $x = 1e-2$ stalls at $+2.46e-9$, the deep path wants care; the tightness — every measured point above $\sqrt{\lambda^*}$, the floor within $1.1e-11$ of it, so the compression infimum is consistent with $\sqrt{\lambda^*}$ EXACTLY: TIGHT. The power-sum class does not dip below $\sqrt{\lambda^*}$ at any measured level; the shadow-equivalence theorem's target is confirmed sharp on the free side, with the abelian side's $D_{\text{abelian}}(2) = \sqrt{\lambda^*}$ in the closure sense from the corrigendum chapter. And the B-exit bookkeeping closes the geometry: the wall's ROOT box caps B at 120, so the family segment $x < 1.65e-3$ — B beyond 120 — lies outside the certified box, covered analytically by the parameter-space-global compression bound. The growth laws beyond every box are the one residue this leaves named, and the ledger chapter carries it.

The critical locus is closed. The unboundedness is a parameter-space artifact that cancels in value; the tightness at $\sqrt{\lambda^}$ is the supported structure, measured to $+1.1e-11$ and certified on the plateau; and the compression theorem stands as the free side's sharp lower bound. The shadow-equivalence theorem's target is confirmed sharp on both sides.*

The law	The statement	The certificate
THE PLATEAU LAW	the line-atom family's kernel saturates at $\sqrt{\lambda} + 2.1491e-9$ ($x \leq 1e-3$, $K \geq 36$): the valley is unbounded in B ($B = c/2x$) but FLAT in value — the $I=1$ stripe is the x^0 leading structure, the B -magnitude cancels identically	CERTIFIED at mpmath precision 120, $K = 64$: 1.2771421150615995710039
THE APPROACH LAW	$E(x) - \text{plateau} \sim x^{4.00}$ — the stationary family's generic quartic approach onto the plateau	R^2 1.0000 (the log-log fit)
THE K-TAIL LAW	$ E_K - E_{\infty} \sim \exp(-0.805 K)$ — the y^K geometric truncation tail; the $K \geq 36$ floor below $1e-12$	the K -chain measured, the floor certified
THE OFF-FAMILY FLOOR	the descent at the converged $K = 64$ lands $+1.132e-11$ above $\sqrt{\lambda}$ — 60x deeper than the Task-35 winner ($+6.4e-10$), driven by tiny displacements (the B 's $\sim 4e-6$, the couplings $\sim 1e-7$, the C 's $\sim 3e-8$); the descent from $x = 1e-2$ stalls at $+2.46e-9$ — the deep path delicate	the SVD backward error $\pm 6.3e-13$
THE TIGHTNESS STATEMENT	every measured point sits ABOVE $\sqrt{\lambda}$ and the floor descends to within $1.1e-11$ — the compression infimum is consistent with $\sqrt{\lambda}$ EXACTLY (TIGHT); the power-sum class does not dip below $\sqrt{\lambda}$ at any measured level	the compression theorem honored at every rung of the ladder
THE B-EXIT	the wall's ROOT caps B at 120: the family segment $x < 1.65e-3$ ($B > 120$) lies outside the certified box	covered analytically — the parameter-space-global compression bound

Table 8. The critical locus's structural law, six parts: the plateau, the approach, the K -tail, the off-family floor, the tightness, and the bookkeeping. Together they close the residue named at Task 35 — the line-atom approach's thin valley, $B \sim c/2x$ unbounded — by showing the unboundedness is a parameter-space artifact that cancels in value.

8 · The Regenerated Ledger — the Batteries, the Dictionary, the Residue

The corrigendum arc's instruments are eight, and Table 9 rows them the way the first edition rowed its one: each battery, its delivery, its certificate tier. The arc they compose is the programme's whole method in miniature — the adjudication that retracts an error (Task 32), the engine that replaces a wall with a measured structure (Task 33), the formalization that certifies the structure (Task 34), the drain that turns the certificate into coverage (Task 35), the second wall built and closed the same way (Tasks 35 and 36), the cost law that explains the drain's remaining expense (Task 38), the controlled experiment that kills a banked claim before it ships (Task 39), and the closed form that finishes the mathematics (Task 40). The README's battery table carries these rows alongside the first edition's; the results JSONs carry every number this record quotes; the reading order is unchanged.

The battery	The delivery	The tier
abelian_closure.py + abelian_lower_bound.py + h2_exact.py (Task 32)	the class-level adjudication: the boundary valley measured (the monotone descent to the line atom, the anisotropy ~ 888), the exact quartic closure (the charpoly a polynomial in x^4 only), THE ESCAPE RETRACTED; the p-convexity theorem (the box-count wall's dimension halved, $h^6 \rightarrow h^4$ + a convex inner program); kappa_ridge resolved ($+1.3654e-4$ positive, the V-cone, the extraction at precision 120)	MEASURED + machine-exact (the symbolic charpoly; the 400/400 convexity check)
abelian_cover4d.py (Tasks 33-35)	the 4-D covering: the true-norm 6x6 pencil (hole-free, validated to $8e-15$), the BDC corner trio, the DPP patch mode (1804 patches, median r $4.23e-2$ — the 11x coarser certified boxes), the orbit reduction (16 quadrants = 6 representatives); the drain at 55.5M calls / 27,686,873 leaves / ZERO stalls	CERTIFIED (sound interval bounds on the covered region; $D_{\text{abelian}}(2) = \sqrt{\lambda}$ in the closure sense with part A)
probe_task35.py / probe_task35b.py (Task 35)	the free wall's design probes: the compression theorem measured (the infimum AT $\sqrt{\lambda}$), the multi-start winner IS the line-atom approach), the tube-lift composition KILLED (the true transverse curvature not uniformly positive, $-1.67e+4$), the window's far-field explosion ($x_2 \rightarrow 1.78$, $x_8 \rightarrow 43.0$)	MEASURED (the reproduction-first discipline)
free_class_wall.py (Tasks 35-36)	the 12-parameter wall's engine — the cheap-first chain (the window, the e0-poly, the powered Gershgorin, the trace-power far-out test, the full Rayleigh) — and the rho-boundary closure: the word-power recursion, the partial-sum sandwich; the wall at 720k calls / 316,401 certified / ZERO stalls	CERTIFIED ($D_{\text{free}} \geq \sqrt{\lambda}$ on the certified region)
probe_task36.py / probe_task36b.py (Task 36)	the unstable-mode forms' probes: the zero-WFA anchor EXACT ($0.00e+00$ over 20 randoms, ρ 1.26-5.79), the near-locus divergence chain ($33.8 \rightarrow 7.9e7$), the X-cancellation conic's convergence at $\rho > 1$, the excited growth law $\rho^{(2N)}$, the coverage $1755/2000 = 87.8\%$ at $N = 2$	MEASURED + EXACT anchors
tail_law.py (Task 38)	the tail's structural law, five parts: the fuel law ($m \sim \rho^{31.46}$, $R^{2.0974}$), the race law ($d^* = \text{ceil}(\log_2(\text{eps}_0)/\text{rate})$, median $ \text{err} $ 1.0 level), the $N^* = 2$ law, the censored = slow-not-stuck, the gradient carriers	MEASURED (the race law predictive to 1 level)
gradient_split_ab.py (Task 39)	the gradient-split corollary REFUTED by the controlled A/B on the live stack: the stock 0.4998 vs the gradient 0.4185 e45/call (0.84x, 324 cap-hit stalls); the carrier profile a transient; the width-first rule validated near-optimal ($0.135 \approx \log_2(100/91)$ bits/level)	MEASURED (the negative result, banked before any engine edit)
critical_locus.py (Task 40)	the critical locus closed: the closed-form kernel (validated against the general machinery to 0.0), the structured $O(n)$ matvec (vs the dense SVD, $4.4e-16$), the plateau certified at precision 120, the off-family floor $+1.132e-11$, the tightness statement	CERTIFIED (mpmath precision 120, $K = 64$)

Table 9. The corrigendum arc's battery rows: eight instruments, Tasks 32 through 40, each with its delivery and its certificate tier. The rows live in the repository's README battery table alongside the first edition's; the results JSONs carry every number this record quotes.

The dictionary — the corpus's running index of its own laws — gains six entries from the arc, and they are the volume's most portable content. THE BOUNDARY-VALLEY LAW: a reported comparison between two classes' optima is a claim about infima, and a scan's best point is not an infimum — anisotropic valleys stall optimizers at points that then masquerade as distinct basins; the detection instrument is the reproduction-first discipline plus a structure probe on the near-optimal coordinates. THE CORNER-IDENTITY LAW: a polynomial corner form evaluated at a fixed test vector can certify at coarser resolutions than any eigendecomposition instrument, provided its cross-terms are

poison-free — the eigendecomposition is the expensive way to the same bound. THE WORD-POWER LAW: a divergent operator's partial sums, when PSD-monotone, certify the divergence's lower bound without convergence — the polynomial sandwich replaces the convergence gate wherever the class vectors make the denominator constant. THE RACE LAW: an interval swamp is a radius problem first — predict the conversion depth from the radius-over-value and the per-level decay, and census failures become budget line-items rather than walls. THE TRANSIENT-PROFILE LAW: a measurement-time profile of a live state is an artifact of that state until re-measured — banked priors on a moving frontier go stale, and the controlled A/B is the cheap instrument that catches it before the edit. THE PLATEAU LAW: a parameter direction unbounded in coordinates can be flat in value to leading order — the x^0 structure cancels the magnitude, and the 'critical' unboundedness is a bookkeeping fact, not an obstruction. Each entry is anchored to this arc's batteries; none of them is a placeholder.

The regenerated open ledger is shorter than the first edition's, and more honest about what kind of work remains. The escape's row is closed — retracted, the equality restored in the closure sense, the target confirmed sharp on both sides. The box-count wall's row is transformed: the wall itself is gone, replaced by the two certificate engines and their continuation grind — the cover4d frontier (28 stack branches at this regeneration, the remaining orbit representatives' p-subtrees) and the wall's far-out refinement (41 stack entries; the FAROUT_CAP = 48 cap-hitters the honest census remainder, slow rather than stuck by the race law's own arithmetic). The unbounded far-field laws remain the one named mathematical item: the window and e0-poly growth laws beyond the ROOT box — the polynomial certificates' monotonicity in the unbounded regime, the B-exit's generalization, the last of the wall's FW-4 residue items. And the seed-level boundary stands unchanged, the empirical face's one open row, waiting on evidence rather than construction. Table 10 is the ledger; the instruction it carries forward is the corpus's standing order — the drain driver, the checkpoint, the certificates' steady accumulation.

The open item	The precise statement	The owner instrument
The continuation grind	the cover4d LIFO frontier (28 branches at this regeneration — the remaining orbit representatives' p-subtrees) and the wall's far-out refinement (41 stack; the FAROUT_CAP = 48 cap-hitters the honest census remainder — the tail is slow, not stuck, per the race law)	the foreground drain driver (the checkpoint/resume protocol; the standing order)
The unbounded far-field laws	the window/e0-poly growth laws formalized — the polynomial certificates' monotonicity in the unbounded regime beyond the ROOT box (the B-exit's generalization: the window's far-field explosion measured at $x_2 \rightarrow 1.78$, $x_4 \rightarrow 9.9$, $x_8 \rightarrow 43.0$)	the wall's machinery (the growth-law battery — the last of the wall's named FW-4 residue items)
The seed-level boundary	the sheaf coboundary orders the out-of-distribution splits exactly at the task level and is null at the seed level — split-level-yes, seed-level-no stands as measured	the Vol IX second edition's experiments (unchanged by this arc)

Table 10. The regenerated ledger: the open items after the corrigendum arc. The escape's own row is closed — retracted, the equality restored in the closure sense, the target confirmed sharp on both sides; what remains is the continuation grind, the unbounded far-field laws, and the seed-level boundary.

Provenance: every statement in this second edition's new chapters is either in the manuscripts it cites (the Corrigendum record through Task 40, the repository's README battery rows, the task batteries' results files, each with its own machine anchor), or derived in-session by the named batteries with their results JSONs as the anchor, or labelled open — and the open ones are exactly the ledger's rows: the continuation grind, the unbounded far-field laws, and the seed-level boundary. The first edition's four chapters stand as the historical record they were certified to be; the corrigendum chapters correct, complete, and supersede their adjudication, in that order, with every correction anchored to the instrument that found it. This volume's place in the reading order is now the long one — the consolidation that found its own error, retracted it, and built the certificates both sides of the equality still stand on — and the file the programme has carried since its first volume closes the record the same way: diagnose the obstruction before choosing the repair, price the repair before claiming it, and certify the price.